ESAP-SPHYNX-026

ER41XX SDK_EZ User Guide

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Revision History

Rev	Date	Description of Change	Approved
1.0	2026/03/16	Initial release	
2.0	2026/09/29	Detailed description of the new configuration tool EzToolkit	
2.1	2026/11/09	▪ **Corrected figure numbering and figure references;** ▪ **Updated figures:** Figures 1, 2, 4, 5, 8, 11, 13, 14, 15, 17, 18 and 19; ▪ **Corrected misnamed terms:** EzFirm, EzToolkit and UART; ▪ **Corrected descriptions:** FIFO structure, CRC Polynomial, Shutdown (SHD) function, PCR and DVB connection; ▪ **Added Figure 16 and table numbers;** ▪ **Added example descriptions:** GIO_TRBSY, PowerSaving_Shutdown and WakeupTimer(WUT); ▪ **Revised the section title format;** ▪ **Added Section 2.2.3;** ▪ **Added List of Figures and List of Tables**	

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1 *Overview*

SDK_EZ is a system for the ER4100 that makes it easy for developers to write code on resource-constrained microcontroller platforms. As shown in Figure 1, it provides a graphical tool and example code, which run on the PC side and the chip side respectively.

- **EzToolkit:** a graphical tool on the PC side that generates the configuration header file developers need (default: SPI_ER41XX_config.h). The generated file is then copied into EzFirm, overwriting the corresponding file there. In addition, the tool can control the ER4100 demo board (ER4100 DVB) via UART.
- **EzFirm:** examples for the ER4100. Code projects are provided for each function of the chip, and every project can import the configuration file generated by EzToolkit so that the chip behavior meets the development requirements.

1.1 *Operating Architecture and Environment*

As shown in Figure 1, EzToolkit, together with the DVB evaluation board and EzFirm (example code), covers RF parameter configuration, register generation, TX/RX testing, frequency offset calibration, and finalizing mass-production parameters.

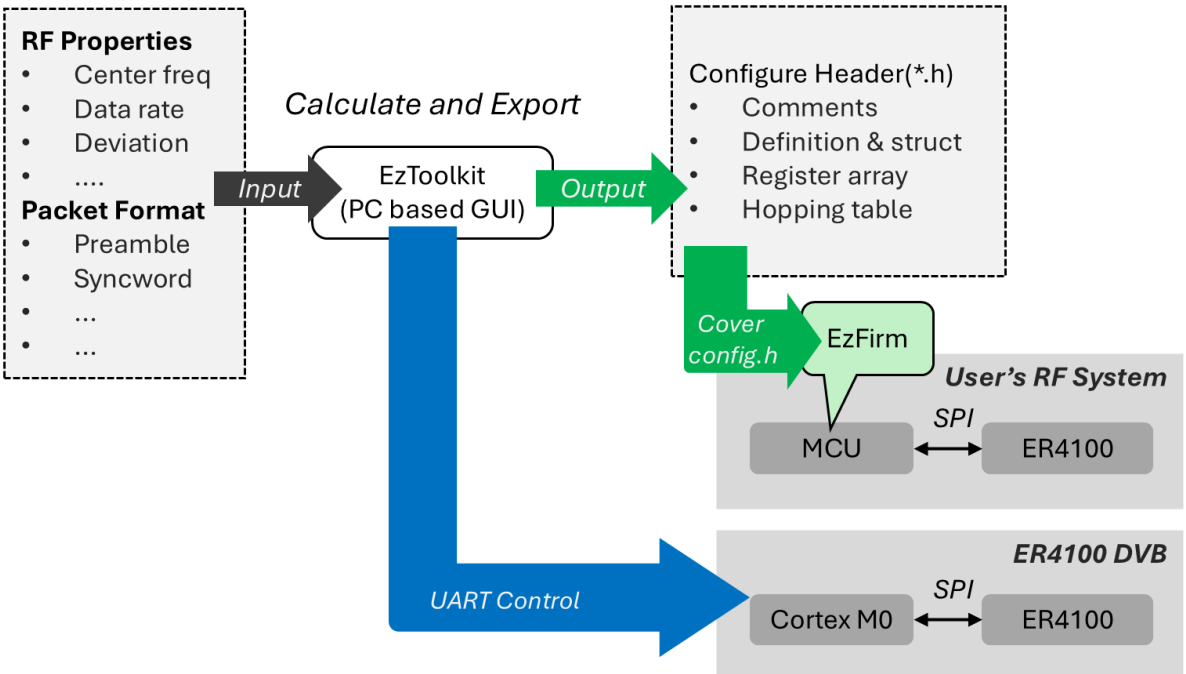


Figure 1. SDK_EZ Architecture

Version and operating environment:

Table 1. Version and Operating Environment

Item	Content
Software version	ER4100_EzToolkit V2.0
Supported chip	ER4100 series SubG RF transceiver
Supported firmware	ER4100_EzFirm V2
Communication interface	USB to serial (UART), DVB evaluation board connection

1.2 *EzFirm_v2 Code Package (8051 Example)*

1.2.1 ***Code List and Summary***

The example applications currently provided by the RF application layer are listed below.

×: not supported by hardware

○: supported by hardware, code provided

▲: supported by hardware, code not provided

Table 2. Code List and Summary

Packet communication project	MAC format		Trigger interface	Implementation summary
	Transparent PreSync	802.15.4		
NFC_Field_Interrupt	don't care		UART	Waits for a phone to approach using an interrupt
NFC_Field_Polling	don't care		UART	Polls whether a phone is approaching
NFC_StatusTrigger	don't care		UART	Lets the phone trigger an interrupt via the User Status function in the ER4100 NFC registers
NFC_ReadWrite	don't care		UART	Reads and writes the ER4100 NFC memory from both the phone (NFC) and the host MCU (SPI)
NFC_Menu	don't care		UART	Integrates the NFC functions above, with a UART menu for developers to test and verify

Power Saving	don't care		P05	Sleep and wake-up operations; the state is toggled via P05.
Power Saving Shutdown	don't care		P05	Puts the chip into Shutdown (SHD) mode via the ER4100 GPIO1 SHD pin (lowest power); P05 toggles Shutdown mode and power-up, and the chip is re-initialized after power-up. The SHD function must be enabled in PCR first
Power Saving WUT	don't care		P05	The wake-up timer periodically wakes the sleeping chip
Wakeup Timer	don't care		UART	Runs the wake-up timer while the chip stays active; the period and mode (periodic or one-shot) are set via UART, and timer events are output
Single Tone	don't care		P05	P05 triggers the chip to start or stop transmitting a continuous carrier
RSSI Scan	don't care		None	Samples the energy of the current channel and outputs the dBm value via UART; can be tested together with Single Tone
Standard	○	○	P05	Keeps RX listening continuously; after a packet is received, the data and RSSI are output via UART. Pressing the button (P0.5) manually triggers a TX transmission.
Standard with header and trailer	○	X	P05	Adds a packet header and trailer mechanism in Transparent mode Header : length and packet number Trailer : CRC
Long Packet	○	X	P05	Long packet mode: moves the FIFO data in segments to overcome the FIFO length limit (128 bytes)

WOR	○	▲	P05	Wake-on-receive (WOR): periodically opens a short RX window during sleep. After a packet is received, the MCU wakes up to read the data and, once processing is complete, re-enters the WOR cycle
WOR + ack	○	▲	P05	Adds a short acknowledgment packet to the WOR flow This application checks the length, packet number and CRC
CCA	▲	○	P05	Same flow as Standard, but clear channel assessment (CCA) is performed before TX to confirm ambient interference is low before transmitting
Addressing	X	○	UART	Implements addressed packet transmission and reception under 802.15.4 (triggered by UART characters)
PER Test	○	○	P05	Based on the Standard project: the transmitter sends packets continuously while the receiver accumulates and calculates the packet error rate (PER)
GIO_TRBSY	○	▲	P05 / UART	Based on the Standard project: outputs the TX/RX busy status (TR_BSY) to ER4100 GPIO0; UART commands switch the GPIO0 mode and turn RX on or off

UART pin assignment (UART0, 115200 baud):

- TXD : P06
- RXD : P07

1.2.2 *Directory Structure*

ER4100_EzFirm

```
├── Common/        # Shared utilities: delay, UART, common type definitions
├── Include/       # MCU system header files (N76E003.h, SFR_Macro.h, etc.)
├── RF Drivers/    # RF application layer, HAL layer and ER4100 SPI API
├── User/          # User application entry (main.c)
├── Project/       # Keil uVision project files and build output (*.hex)
```

1.2.3 *Software Architecture*

The current architecture of EzFirm is shown below. The example applications write the configuration file contents to the chip through the programming interface. If developers use a different MCU, they need to implement the SPI and GPIO functions themselves.

Figure 2 shows the simplified architecture of EzFirm. Developers generate the configuration file according to their requirements, then modify or directly call the functions of the example application layer as needed. If an MCU other than the one used by the example code (N76E003) is used, the MCU's SPI and GPIO implementation must also be modified.

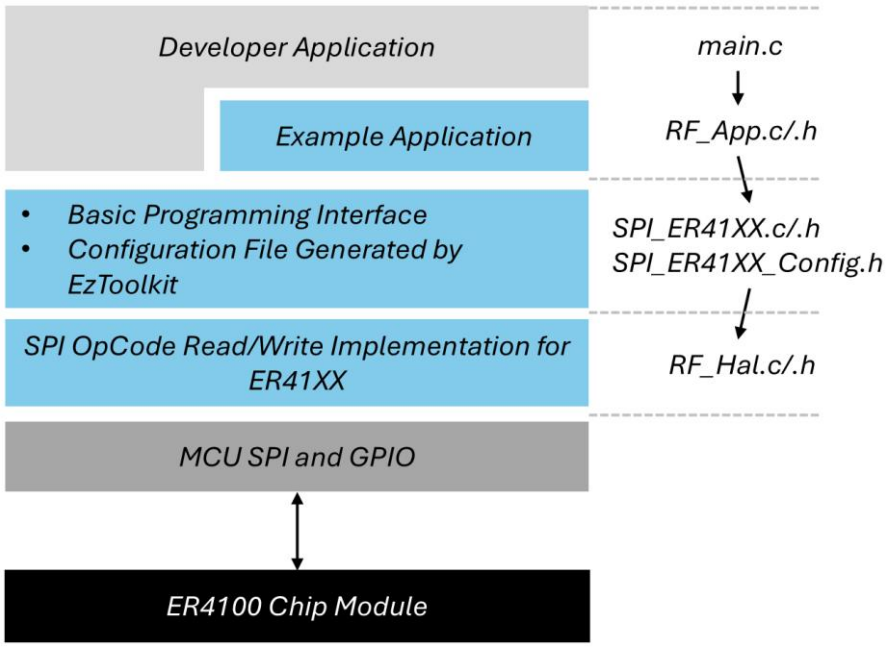


Figure 2. EzFirm Code Layer Architecture

- Main application layer: main.c
  Initialization, main loop
- RF application layer: RF_App.c / RF_App.h
  Packet TX/RX, IRQ handling, RX buffer management
- RF lower-level application interface: SPI_ER41XX.c / SPI_ER41XX.h ← SPI_ER41XX_config.h (register configuration)
  Register read/write, FIFO operations, mode switching
- RF SPI access layer: RF_Hal.c / RF_Hal.h
  SPI timing (software bit-bang), pin definitions

  Calls TRx_Write / TRx_Read to perform the actual SPI transfers

1.3 *Overview of the Main Window Tabs*

Figure 3 shows the main operating window of the software. It has three main areas: Parameter (parameters), DVB Tester (DVB debugging) and REG Table (register table), described below.

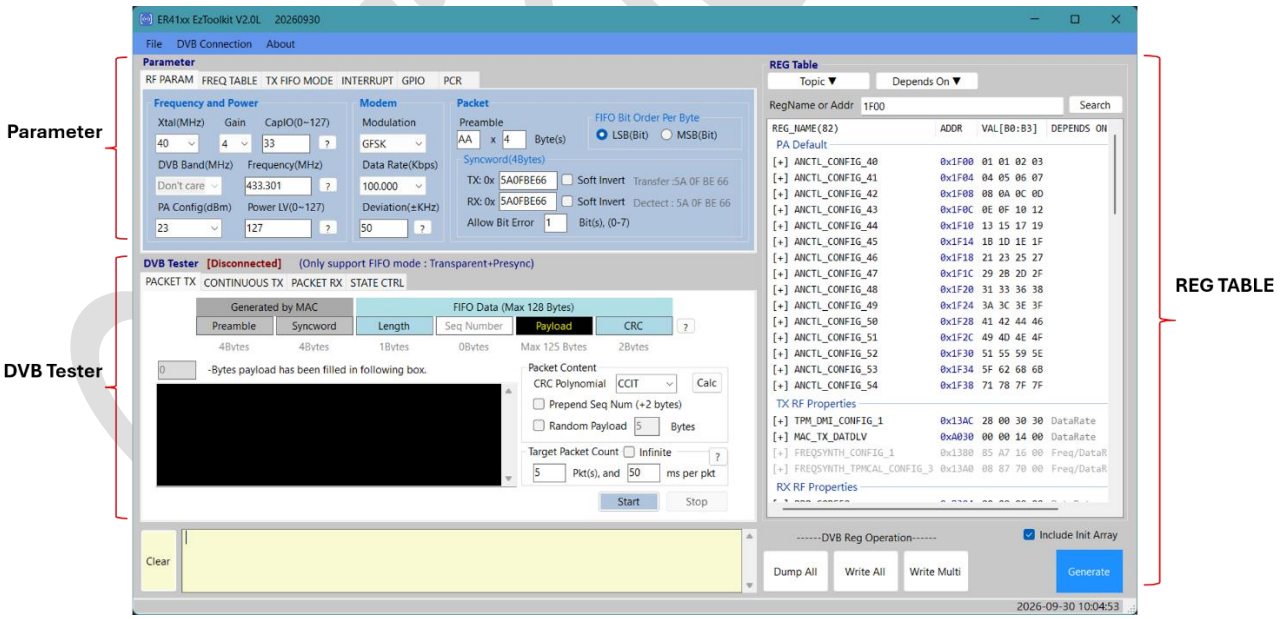


Figure 3. EzToolkit Operating Window

Parameter-Configuration Panel

- RF PARAM — core RF parameter configuration (crystal, frequency, power, modulation, frame format)
- FREQ TABLE — management of the predefined channel frequency table
- TX FIFO MODE — selection of the TX FIFO operating mode
- GPIO — pin function configuration
- PCR — power management and reset control

DVB Tester-Test Panel

- PACKET TX — single-packet transmission test: sets the FIFO contents (preamble/length/sequence number/Payload/CRC); supports random Payload and CRC calculation; the packet count, interval or Infinite mode can be set; one-click Start/Stop
- CONTINUOUS TX — continuous wave/pattern transmission test: SingleTone or Pattern (Bit_01010101, PRBS9) transmission modes; commonly used for power and spectrum measurements
- PACKET RX — single-packet reception test: starts scan reception, counts OK/Err packets and RSSI, and provides a reception Log and packet content comparison
- STATE CTRL — chip power state switching; provides Idle / DeepSleep two-state control

REG Table-Register Table View

- DVB Reg Operation – DVB register read/write buttons; after selecting multiple rows, the selected registers can be written with the Write Multi button
- Generate – generates a macro definition file based on the view, for use in EzFirm

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2 *Parameter-Configuration Panel*

2.1 *RF PARAM-Core RF Parameter Configuration*

This section describes the core RF parameters in the parameter area (RF PARAM). As shown in Figure 4, the blocks from left to right are:

- **Frequency and Power :** crystal configuration / frequency and band configuration / TX power configuration
- **Modem :** modulation type / data rate / frequency deviation
- **Packet :** preamble / sync word / FIFO transmission bit order per byte

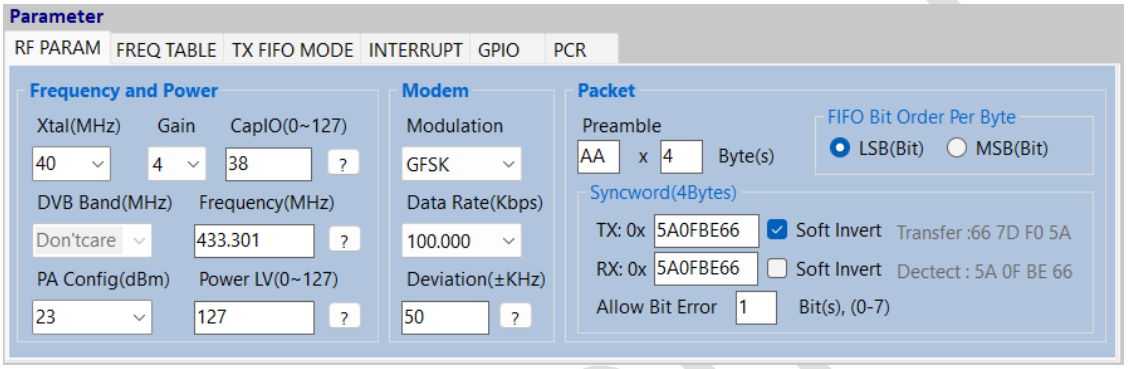


Figure 4. Main Window Parameter Area – RF PARAM Tab

2.1.1 *Crystal Configuration (Xtal / Gain / CapIO)*

Supported crystal options:

38.4 MHz — corresponds to the DR_xtal38.4m data rate table
40.0 MHz — corresponds to the DR_xtal40m data rate table (current default)

After a crystal is selected, the Datarate drop-down list automatically switches to the data rate options supported by that crystal. The available data rate range differs between crystals.

The crystal selection determines the chip reference clock, start-up characteristics and RF frequency offset accuracy, and is the basis for proper RF communication.

- **Xtal (external crystal frequency)**: must match the crystal actually soldered on the hardware (commonly 40MHz). All chip frequencies and baud rates are derived from this crystal; a wrong setting directly causes frequency offset, severe packet loss and communication failure.
- **Xtal Gain (crystal start-up gain)**: controls the crystal start-up speed at power-up/wake-up and the static power consumption

  Gain=4: typical current 1.8mA; recommended default, balancing power consumption and start-up speed

  Gain=8: typical current 2.4mA; faster start-up, higher power consumption

  Engineering rule: default is 4 at room temperature; increase only if crystal start-up is unstable at low temperature
- **CapIO (load capacitance trim)**: RF frequency offset calibration parameter. Trimming the crystal load capacitance corrects the frequency offset error of the whole device. Once the PCB design is finalized, calibrate with a spectrum analyzer and fix the optimal value to ensure carrier frequency accuracy.

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2.1.2 *Frequency and Band Configuration (Frequency / DVB Band)*

2.1.2.1 *Chip-supported bands*

The ER4100 hardware supports four Sub-G bands, fully covering the mainstream ISM bands:

99 ~ 125 MHz
196 ~ 253 MHz
391 ~ 508 MHz (includes the 433MHz ISM band)
780 ~ 1018 MHz (includes the 915MHz ISM band)

2.1.2.2 *Parameter definitions*

- **DVB Band**: only an identifier for the DVB evaluation board test tool. It is determined by the DVB hardware daughter board and cannot be changed in software. It is only used to filter the PA power options in the UI, **does not restrict the actual RF frequency of the chip**, and has no effect in mass-production firmware.
- **Frequency:** RF carrier reference frequency, which can be set freely within the bands supported by the chip. Note: a frequency supported by the chip ≠ one supported by the PCB matching network; the frequency setting must match the hardware circuit design.

2.1.3 *TX Power Configuration (PA Config / Power LV)*

Complete PA Config option list (confirmed by binary measurement):

Table 3. PA Config Options

Type	Band	Nominal power	Power LV range
H-PA (high power)	433 MHz	23 dBm	0~127
H-PA (high power)	915 MHz	23 dBm	0~127
L-PA (low power)	433 MHz	10 dBm	0~27
L-PA (low power)	868 MHz	14 dBm	0~27
L-PA (low power)	915 MHz	10 dBm	0~27

Note: PA Config options are filtered dynamically according to the DVB Band selection, and only the power options for the current band are shown. The hardware matching network design determines the available options, and the software selection must match the PCB design.

Power configuration is strictly bound to the hardware PA circuit. Software cannot switch the chip's PA type; it only matches parameters and adjusts amplitude.

- **PA Config (power option selection):** corresponds to the hardware H-PA (high power) / L-PA (low power) matching network and must match the PCB design. If the hardware is L-PA, forcing the software to 23dBm will cause an impedance mismatch, abnormal output power and PA overheating, which may damage the device.

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- 23dBm: corresponds to the H-PA high-power circuit
  10/14dBm: corresponds to the L-PA low-power circuit
- **Power LV (fine power level)**: adjusts the TX power within the selected PA option
  To reduce power in a project: keep PA Config unchanged and simply lower the Power LV value.
  H-PA(23dBm): value range 0~127; 127 is full power output
  L-PA(10/14dBm): value range 0~27

2.1.4 *Modulation and Frame Parameters (Packet)*

As figure 4, the RF uses proprietary GFSK modulation, suitable for typical Sub-G low-data-rate communication:

- **Modulation**: GFSK (Gaussian frequency-shift keying)
- **Over-the-air data rate**: 100Kbps
- **Frequency Deviation**: 50kHz (standard ratio: deviation ≈ data rate/2)
- **Preamble**: 0xAA, 4 bytes (optionally 8 bytes)
- **Sync word (Syncword):** 0x5A0FBE66 (4 bytes; must be identical on TX and RX; supports Soft invert)
- **FIFO transmission order:** LSB first (optionally MSB first)
- **Sync word error tolerance:** allows 1 bit of sync word error to improve reception tolerance (up to 7 bits)

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2.2 *FREQ TABLE-Channel Table Configuration*

This page is a **predefined channel frequency list**, used for multi-channel switching, channel scanning, etc.

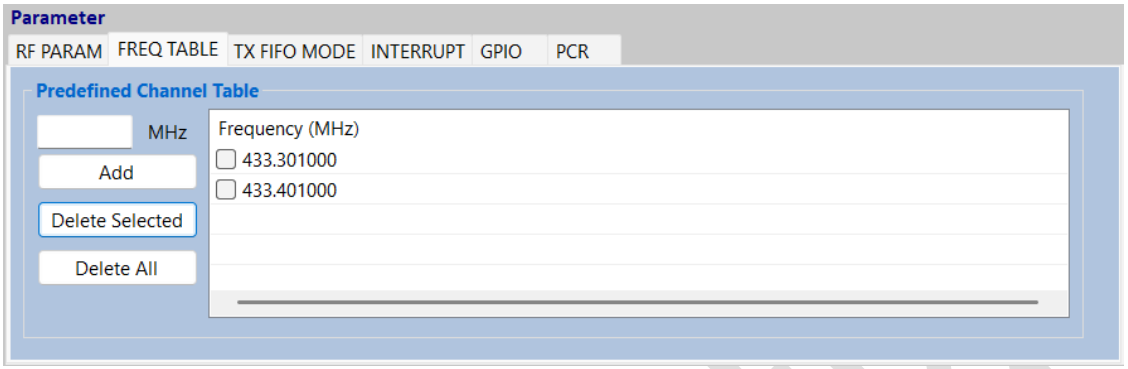


Figure 5. FREQ TABLE

2.2.1 *Function Description*

- Supports batch adding, deleting and clearing of RF frequencies
- Generate: when register code is generated, the channel table is packed into the initialization array as well
- Software can switch frequencies quickly using the channel array list, without recalculating the frequency registers

2.2.2 *Frequency Setting Rules*

- The channel table **stores frequencies only**; all other parameters (data rate, deviation, power, sync word) use the RF PARAM global parameters
- Per-channel RF parameter configuration is not supported; parameters that differ between channels must be configured dynamically in software
- All added frequencies must be within the bands supported by both the chip & PCB matching network

2.2.3 *Channel Table Generation Results*

When Generate is run, the channel table is written to the configuration header file (SPI_ER41XX_config.h). In firmware, call STATUS_TRx TRx_SelectChannel(uint8_t ch_idx) to switch to the specified channel.

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With Crystal set to 40 MHz, the generated channel table is as follows (using 433.301 MHz and 433.401 MHz as an example):

```
/* ------------------------------------------------------------------------*/
/* Hopping Register Data                                                   */
/* ------------------------------------------------------------------------*/
#define NUM_OF_CH_TABLE     2
//====== Register data of channel_pll_table ======
#define CHANNEL_PLL_TABLE_CH0_FREQ {0x08, 0xA6, 0xD3, 0x19}
#define CHANNEL_PLL_TABLE_CH1_FREQ {0xA8, 0x2C, 0xD5, 0x19}

/* ------------------------------------------------------------------------*/
/* Hopping Register Array                                                  */
/* ------------------------------------------------------------------------*/
//====== Array of channel_pll_table ======
#define CHANNEL_PLL_TABLE_ARRAY { \
          CHANNEL_PLL_TABLE_CH0_FREQ, \
          CHANNEL_PLL_TABLE_CH1_FREQ, \
}
```

With Crystal set to 38.4 MHz, the channel table also includes values for registers 0x1998, 0x199C, 0x19A0 and 0x1994, which are written by TRx_SelectChannel (same frequencies as above):

```
/* ------------------------------------------------------------------------*/
/* Hopping Register Data                                                   */
/* ------------------------------------------------------------------------*/
#define NUM_OF_CH_TABLE     2
//====== Register data of channel_pll_table ======
#define CHANNEL_PLL_TABLE_CH0_FREQ {0x08, 0xA6, 0xD3, 0x19}
#define CHANNEL_PLL_TABLE_CH0_1998 {0x1B, 0x90, 0x34, 0x00}  // 1998
#define CHANNEL_PLL_TABLE_CH0_199C {0x70, 0x85, 0x34, 0x00}  // 199C
#define CHANNEL_PLL_TABLE_CH0_19A0 {0x5F, 0x24, 0x34, 0x00}  // 19A0
#define CHANNEL_PLL_TABLE_CH0_1994 {0xC5, 0x8A, 0x34, 0x00}  // 1994
#define CHANNEL_PLL_TABLE_CH1_FREQ {0xA8, 0x2C, 0xD5, 0x19}
#define CHANNEL_PLL_TABLE_CH1_1998 {0xC6, 0x9A, 0x34, 0x00}  // 1998
#define CHANNEL_PLL_TABLE_CH1_199C {0x1B, 0x90, 0x34, 0x00}  // 199C
#define CHANNEL_PLL_TABLE_CH1_19A0 {0x0A, 0x2F, 0x34, 0x00}  // 19A0
#define CHANNEL_PLL_TABLE_CH1_1994 {0x70, 0x95, 0x34, 0x00}  // 1994

/* ------------------------------------------------------------------------*/
/* Hopping Register Array                                                  */
/* ------------------------------------------------------------------------*/
//====== Array of channel_pll_table ======
#define CHANNEL_PLL_TABLE_ARRAY { \
          CHANNEL_PLL_TABLE_CH0_FREQ, \
          CHANNEL_PLL_TABLE_CH0_1998, \
          CHANNEL_PLL_TABLE_CH0_199C, \
          CHANNEL_PLL_TABLE_CH0_19A0, \
          CHANNEL_PLL_TABLE_CH0_1994, \
          CHANNEL_PLL_TABLE_CH1_FREQ, \
          CHANNEL_PLL_TABLE_CH1_1998, \
          CHANNEL_PLL_TABLE_CH1_199C, \
          CHANNEL_PLL_TABLE_CH1_19A0, \
          CHANNEL_PLL_TABLE_CH1_1994, \
}
```

2.3 *TX FIFO MODE-Transmit Modes*

RX FIFO mode has no host-software setting; it is configured dynamically by the software at run time via Trx_RX_Trigger(). The DVB evaluation board only supports Transparent+Presync mode. There are three transmit modes, as shown in Figure 6; they are described in detail in the following sections.

2.3.1 *Comparison of the Three Transmit Modes*

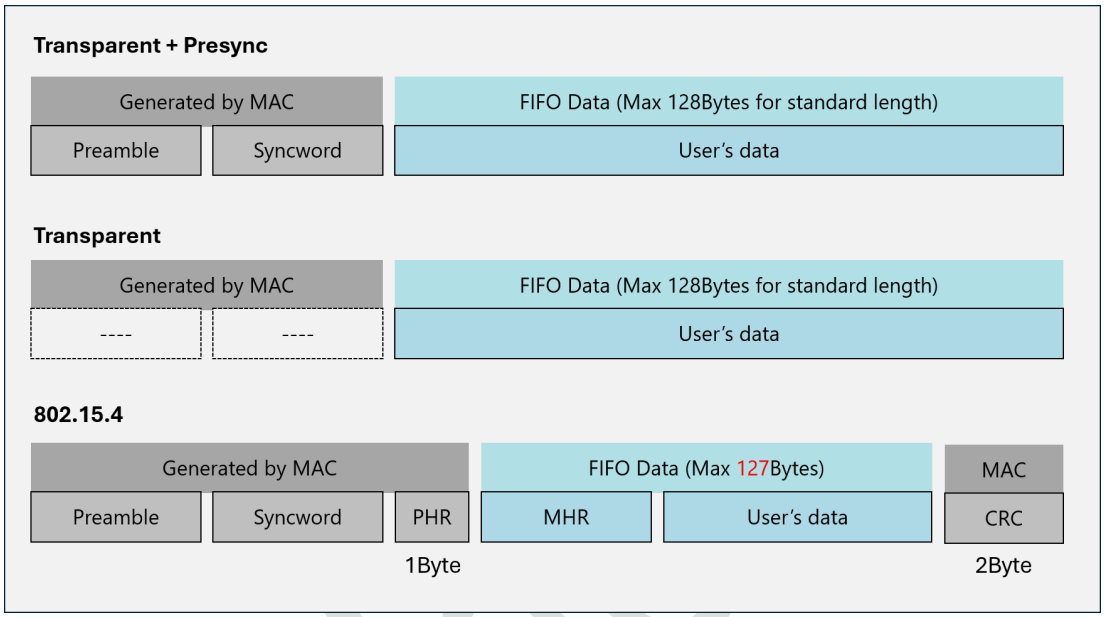


Figure 6. TX FIFO MODE

- **Transparent + Presync (recommended default)**
  - ✓ Hardware automatically prepends the preamble + sync word to the payload
  - ✓ The MCU only writes the payload data; no software frame header assembly is needed
  - ✓ Applicable to: all proprietary Sub-G general communication projects
- **Transparent (pure pass-through)**
  - ✓ FIFO data is transmitted as-is; the hardware does not add any frame structure
  - ✓ The preamble, sync word and frame header must all be assembled by MCU software
  - ✓ Applicable to: custom special frame formats, scenarios requiring full bit-level control
- **802.15.4 Standard Mode**
  - ✓ Hardware automatically adds the preamble, sync word, PHR length field and CRC checksum
  - ✓ The TX data must include the MHR MAC header in the FIFO
  - ✓ Applicable to: standard 802.15.4 protocol scenarios

2.3.2 *Corresponding RX FIFO MODE (Two Modes)*

Although RX FIFO mode has no host-software setting, its operating mode must be set by writing registers in code. Figure 7 shows the two modes. In the physical-layer receive state, the preamble and sync word are checked in either mode, but in 802.15.4 mode the PHR length field and MHR MAC header are also checked. After successful reception, the MHR can still be read from the RX FIFO.

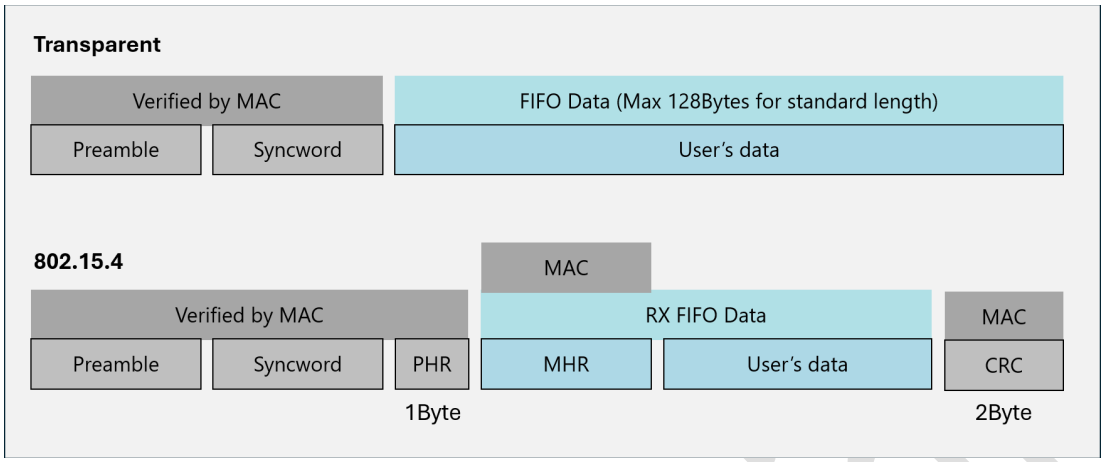


Figure 7. RX FIFO MODE

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2.4 *GPIO-Pin Function Configuration*

2.4.1 *Default Pin Functions*

GPIO configuration items (each pin is configured independently):

Table 4. GPIO Configuration Items

Configuration item	Description
Mode (mode selection)	Selects the pin function: Default_Mode (GPIO0: TCXO_EN; GPIO1: Shutdown (SHD) pin) / GIO_Mode (general GPIO) / internal test modes
GIO Sel (direction selection)	Selects input/output direction in general GPIO mode
Pull Ctrl (pull-up/pull-down control)	Configures internal pull-up/pull-down/floating in input mode
GIO Inv (output inversion)	Whether the output level is inverted in output mode

- **GPIO0**: default TCXO_EN (external TCXO enable), high by default at power-up; if there is no TCXO circuit, it can be reused as a general GPIO
- **GPIO1**: default SHD pin, but it has no effect while the SHD function is not enabled in PCR; once enabled, it takes effect
  - **Pull low: the chip enters Shutdown mode (lowest power consumption)**
  - **Pull high: wakes up the chip**
  - How to enable the SHD function in PCR: set bit2 of register PMU_CTRL_REG (0x0020) to 0

2.4.2 *General GPIO Mode*

If GIO Mode is selected in the GPIO mode option, the following functions can be enabled in the corresponding GIO Select option:

- **TR_BSY :** TX/RX transmission in progress
- **TR_FSH:** TX/RX transmission completed
- **TX_PKT:** TX packet transmission in progress (preamble + sync word + data)
- **TX_DATA:** TX data transmission in progress
- **TX_PKT_O:** TX packet streaming output
- **RX_CD:** RX carrier detection
- **RX_SYNC:** RX data reception in progress
- **FIFO_EXH:** TX FIFO almost empty or RX FIFO almost full
- **TX_DIO_DATA:** TX direct-mode data
- **TX_DIO_CLK:** TX direct-mode clock
- **RX_DIO_DATA:** RX direct-mode data
- **RX_DIO_CLK:** RX direct-mode clock

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- **EVT_WAKEUP:** Wake-up timer interrupt
- **TX_EN :** TX enable signal for controlling an external antenna switch
- **NFC_FIELD_OK :** NFC field detected
- **NFC_BUSY :** NFC operation in progress

2.4.3 *INTERRUPT-Interrupt Events*

When any of the following interrupt events occurs, the INT pin is pulled low until the event is cleared via the registers

- **INT_ST_RX :** a complete packet was received successfully (RX Done)
- **INT_ST_RXERR :** an erroneous packet was received (only valid in 802.15.4 mode; Transparent mode performs no protocol-layer validation)
- **INT_ST_TX :** packet transmission completed (TX Done)
- **INT_ST_TXERR :** an error occurred during transmission (only valid in 802.15.4 mode)
- **INT_ST_TMRTHD :** timer threshold (Timer Threshold) timeout triggered
- **INT_ST_WAKEUP :** woken up from the DeepSleep/PCRMU low-power state
- **INT_ST_NFC :** NFC event triggered
- **INT_ST_TX_FIFO_AEMPTY :** TX FIFO data level below threshold (nearly empty); refill data as soon as possible to avoid underrun
- **INT_ST_RX_FIFO_AFULL :** RX FIFO data level above threshold (nearly full); read out as soon as possible to avoid overflow and data loss
- **INT_ST_RX_SYNCW :** Syncword (sync word) detected, indicating that packet reception has started

2.5 *PCR-Power Management and Reset Control*

The PCR (Power Control Register) page configures the chip's power management behavior and is the prerequisite switch for the Shutdown (SHD) pin function.

Table 5. PCR Configuration Items

Configuration item	Description
PCR Enable	Enables custom configuration of the power management unit Includes: Deep Sleep and Wakeup Timer
PCR Disable	Disables custom configuration of the power management unit
Enable with SHD (TBD)	Same as PCR Enable and includes the SHD function, which allows GPIO1's default mode to act as the SHD pin.

Because the Enable with SHD option is not yet available in EzToolkit, as described in Section 2.4.1, when the chip needs to be controlled via Shutdown mode, a value must be written to the corresponding register to use the SHD pin.

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The following modes are reserved for the chip manufacturer's internal testing; **configuring them in any mass-production project is strictly prohibited**, as they will cause abnormal operation:

- RX_RF_Mode
- RX_BBP_Mode
- NFC_Mode
- Debug_Mode (low level by default, no functional use)

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3 *REG Table-Register Table View*

- **REG list**: shows the real-time values of all RF registers, grouped by function
- **Generate**: generates a complete C initialization array (including registers + channel table) with one click, which can be imported directly into the MCU project
- **DVB Reg Operation :** DVB register read/write buttons

3.1 *View Filters*

There are two sets of filters above the view that determine which items are displayed:

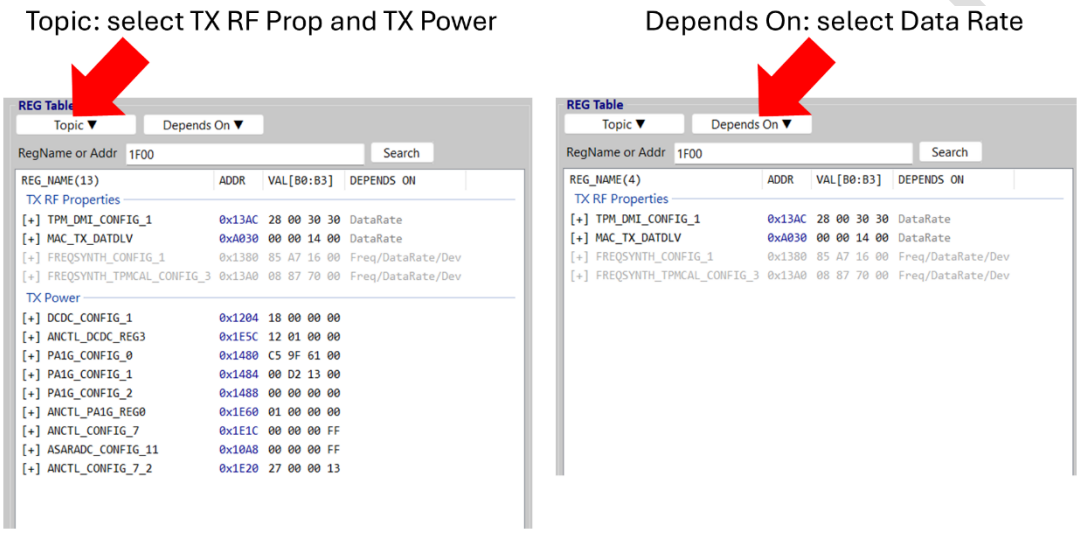


Figure 8. View Filter Usage Example

3.2 *Register Read/Write (with DVB)*

In the DVB Reg Operation area below, the DVB registers can be read and written. In the view, use Shift + left mouse click to select multiple rows, then write the selected registers with Write Multi.

Confi

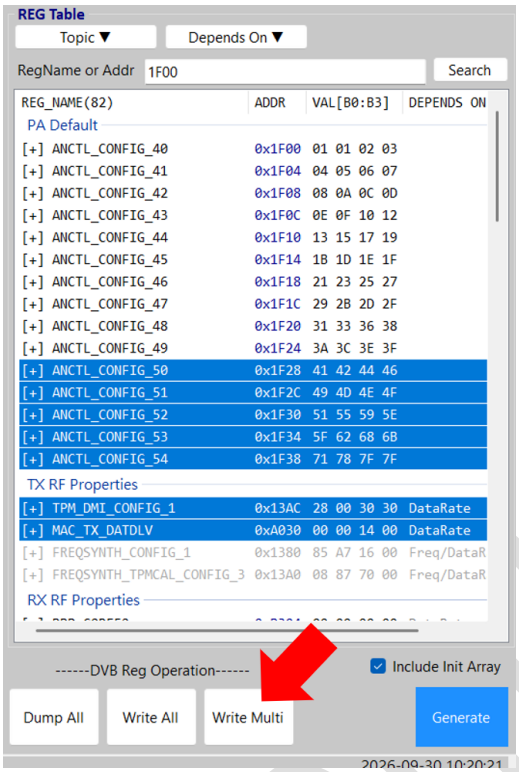


Figure 9. Writing Selected Registers (highlighted rows)

3.3 *Initialization Array Generation*

As shown below, the Generate button generates the initialization array, which is used in EzFirm. Place the generated file in the corresponding path of any project in EzFirm:

ER4100_EzFirm_B_N76_<Example Name>\RF Drivers\ER4100Api

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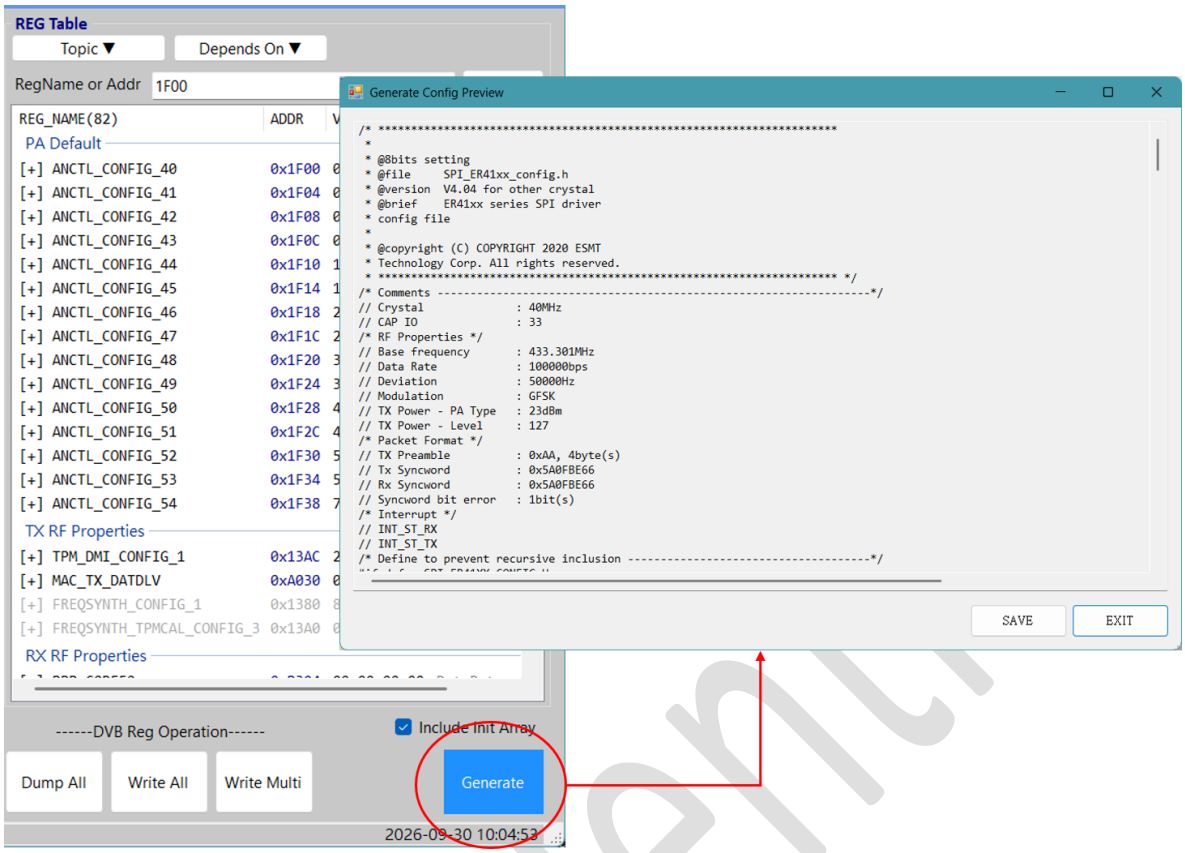


Figure 10. Generate Button

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4 *DVB Tester-Test Panel (Available After Connecting to DVB)*

DVB Tester requires the DVB evaluation board for debugging, as shown in Figure 11.

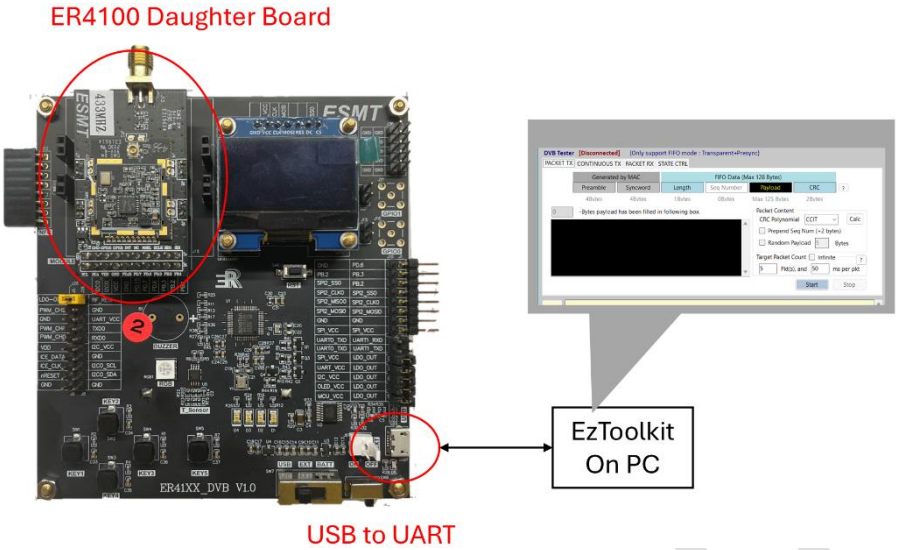


Figure 11. DVB and DVB Tester Diagram

DVB Tester uses a sub-tab structure with 4 test modes, as listed in Table 6.

Table 6. DVB Tester Sub-tabs

Sub-tab	Function	Options
Continuous TX	Continuous transmission test	SingleTone (single tone) / PRBS9 (pseudo-random sequence) / Pattern (fixed pattern)
Packet TX	Packet transmission	Single packet / continuous transmission; FIFO mode selection
Packet RX	Packet reception	Reception count statistics; supports a limited number of packets / unlimited reception
State Control	Chip power state control	Idle / DeepSleep

This guide summarizes the ER4100 host software's page functions, parameter setting constraints and engineering usage rules, including RF parameters, frequency channel table, FIFO transmit modes, GPIO functions and register operations. It serves as a reference for tuning ER4100 register settings and proprietary Sub-G 433M/915M communication.

The DVB evaluation board only supports testing and verification of some ER4100 functions. For the other functions supported by the ER4100, testing based on EzFirm is recommended.

The DVB Tester workflow is described in the following subsections.

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4.1 *Connecting to the DVB via UART*

Open the DVB Connection panel, select the port corresponding to the DVB, check " Write Config to DVB after connection" and click Connect. After the connection succeeds, the tool automatically writes the REG Table contents to the DVB to synchronize them (see Figure 12 and Figure 13; shown in the log).

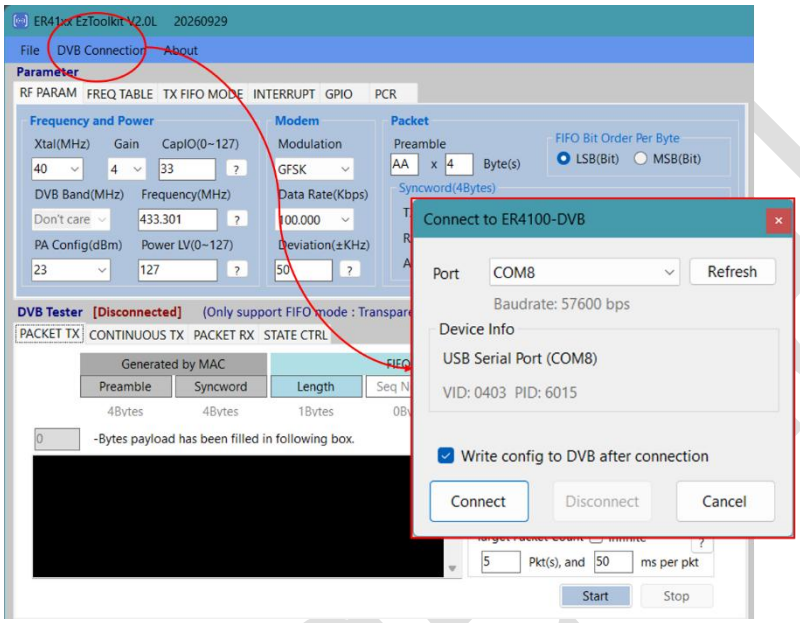


Figure 12. DVB Connection Panel



Figure 13. Automatic Write of the REG Table to the DVB After Connection

4.2 *Configuring Parameters and Debugging*

After connecting, configure the parameters in order and write them to the DVB; testing can then begin in the DVB Tester area.

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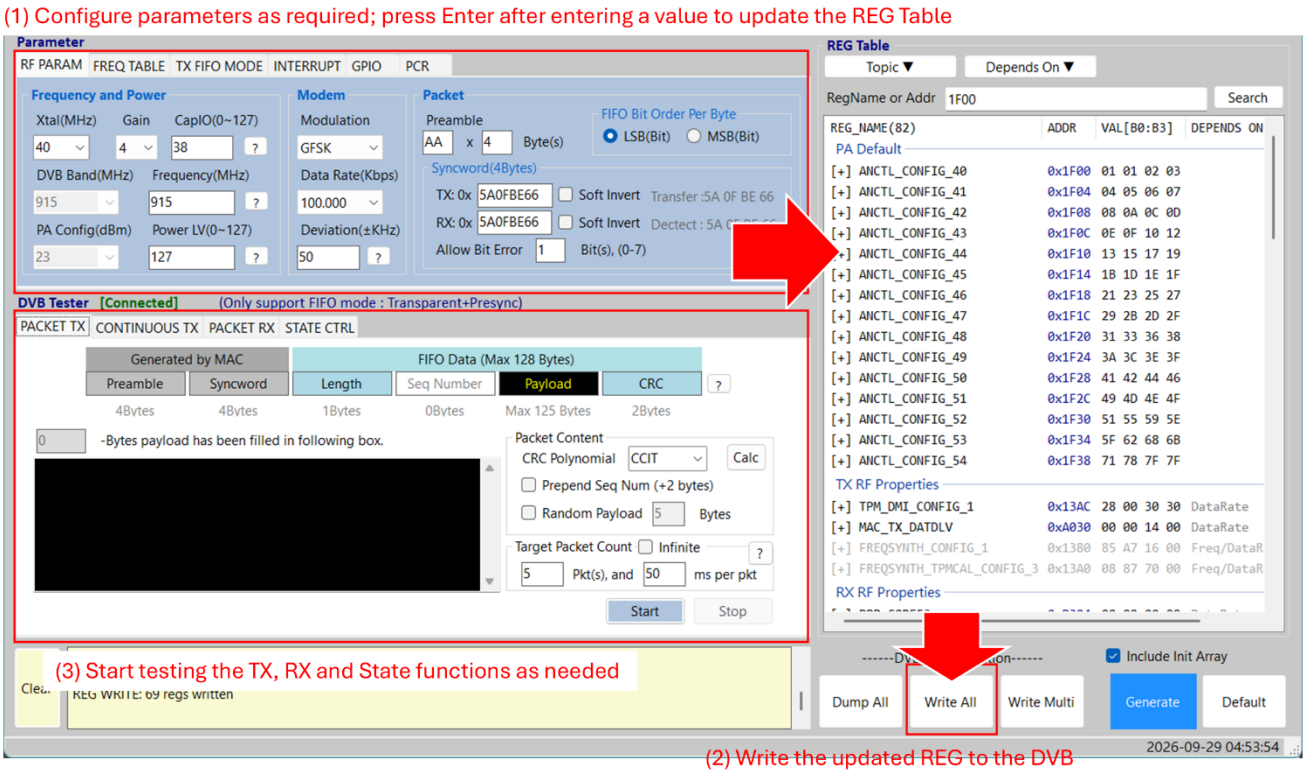


Figure 14. Flow for Configuring Parameters and Debugging

4.3 *PACKET TX-Packet Transmission*

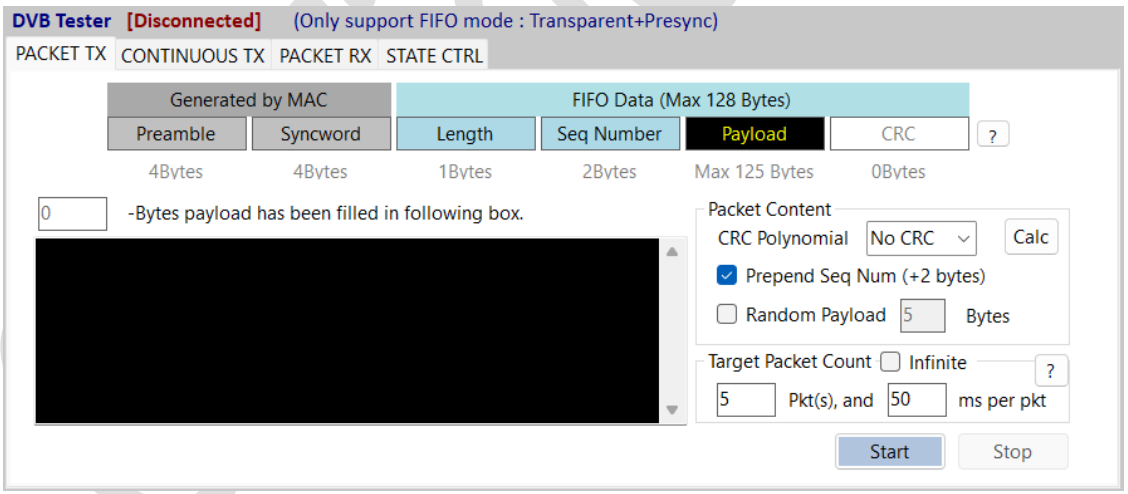


Figure 15. PACKET TX Interface

Apart from the preamble and sync word, the packet length (Length) is always included in a transmission. In addition, the user can set the following parameters:

- CRC Polynomial – selects the CRC polynomial for the transmitted packet, or selects no CRC
- Prepend Seq Num – whether a packet sequence number is prepended to the Payload
- Random Payload – generates a random Payload; if unchecked, the user must enter the Payload (hexadecimal) in the black box on the left
- Target Packet Count – sets the number of packets sent in each round

4.3.1 *Data Format*

The preamble (4 bytes/8 bytes) and sync word (4 bytes) are generated and inserted automatically by the MAC layer — **they are not part of the data you write to the FIFO**.



Figure 16. FIFO Data Structure

All data after the sync word (Length, Seq, Payload and CRC) is limited by the FIFO, with a maximum total length of 128 bytes. Its structure is as follows:

Length (1 byte) + Seq (2 bytes, optional) + Payload + CRC (2 bytes, optional)

Notes:

1. Length field = total number of bytes of (sequence number, if enabled + payload + CRC, if enabled). It does not include the length field itself, the preamble or the sync word.
2. Maximum Payload = 128 - 1 (Length) - Seq (2 if enabled) - CRC (2 if enabled): 123 bytes with both enabled, 125 bytes with one enabled, 127 bytes with neither enabled.
3. CRC, if enabled, is calculated only over the payload and the sequence number (if enabled); **it does not cover the length byte, preamble or sync word**.

4.4 *CONTINUOUS TX-Continuous Transmission*

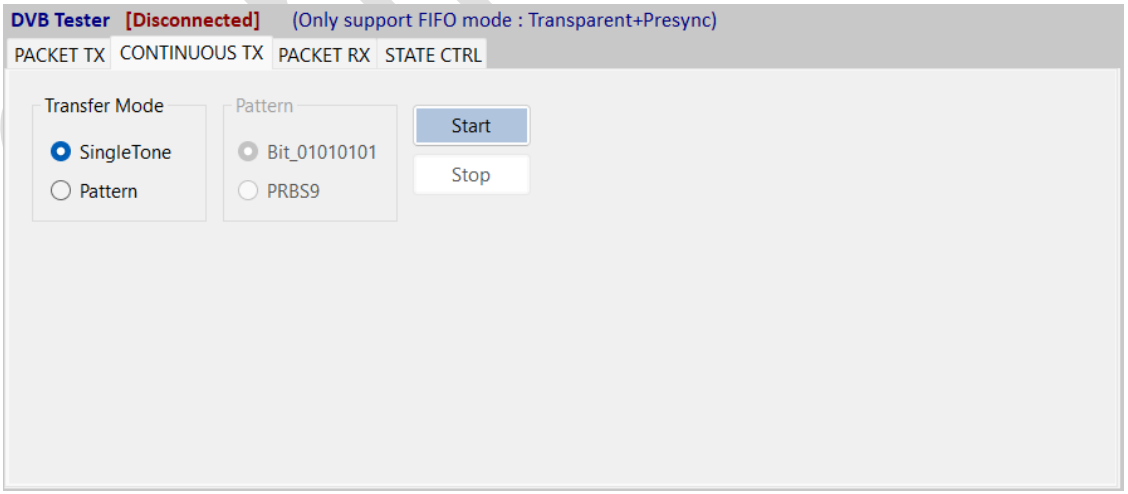


Figure 17. CONTINUOUS TX Interface

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Continuous transmission can be either unmodulated carrier transmission (Single Tone) or continuous data transmission (Pattern)

4.5 *PACKET RX-Packet Reception*

The PACKET RX parameters correspond to those of PACKET TX; in Packet Parser, select the matching parameters. In addition, the Scan RSSI button can be used to measure ambient power only; this function can be used together with CONTINUOUS TX for testing.

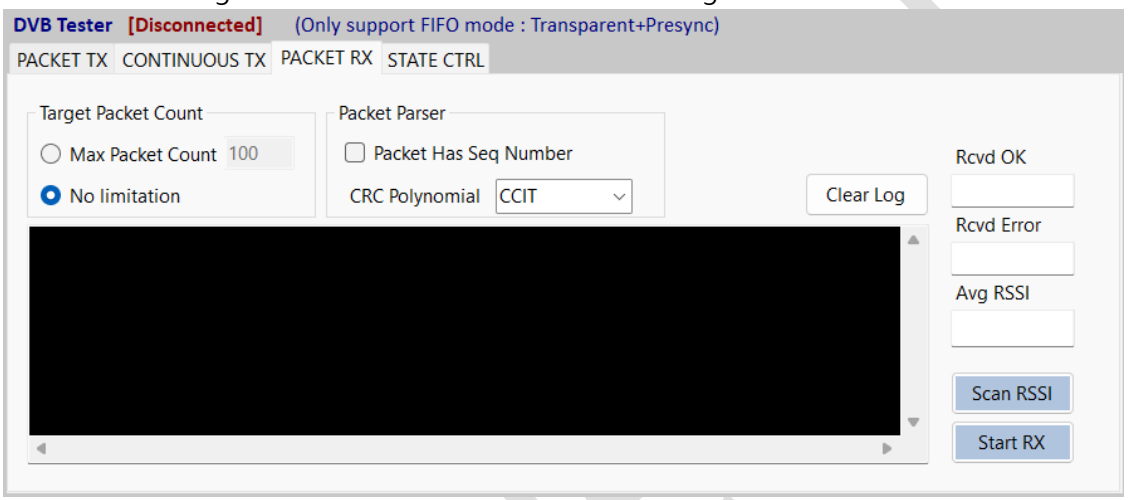


Figure 18. PACKET RX Interface

4.6 *STATE CTRL-Power State Control*

This interface controls the power state of the chip on the DVB daughter board (Idle / DeepSleep).

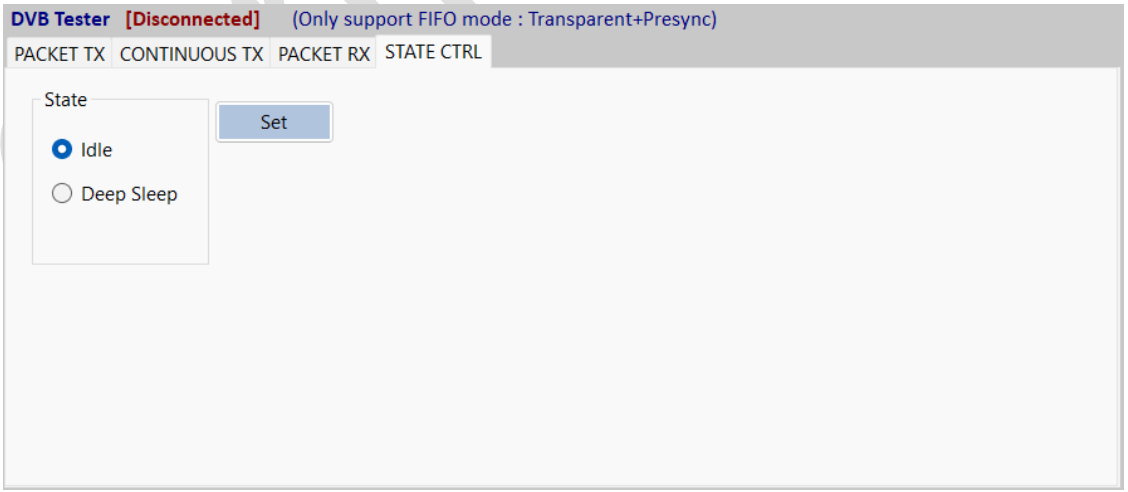


Figure 19. STATE CTRL Interface

5 *Standard Project Configuration Template (433.301MHz)*

Tuned parameters provided as a project reference, which can be reused directly:

- Crystal: 40MHz, Gain=4, CapIO=33
- Carrier frequency: 433.301MHz
- Modulation: GFSK, 100Kbps, 50kHz deviation
- Power: PA Config=23dBm, Power LV=127 (full power H-PA)
- Frame structure: 4-byte 0xAA preamble + 4-byte sync word 0x5A0FBE66, LSB first
- Transmit mode: Transparent + Presync
- GPIO configuration: GPIO0 and GPIO1 are both in their default state, i.e., GPIO0=TCXO_EN and GPIO1=no function
- PCR configuration: Enable

Note 1: PCR Enable does not include the Shutdown (SHD) function. If the SHD function is enabled on the PCR page, i.e., bit2 of register 0x0020 is set to 0, the default state of GPIO1 becomes the SHD pin (high level = working mode, low level = Shutdown mode)

Note 2: The SHD function cannot yet be selected in EzToolkit, but it can still be set by writing the registers directly in the REG Table area

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6 *Key Engineering Notes*

1. PA configuration must match the PCB hardware matching network; configuring beyond the hardware's power option in software is prohibited
2. CapIO is the key frequency offset calibration parameter and must be calibrated with a spectrum analyzer and fixed before mass production
3. The GPIO1 Shutdown (SHD) function depends on the PCR configuration; both are required
4. All internal test GPIO modes are prohibited throughout mass production
5. The channel table stores frequencies only; parameters that differ between channels must be configured dynamically in software
6. DVB Band is used only for evaluation board debugging and can be ignored in mass production

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